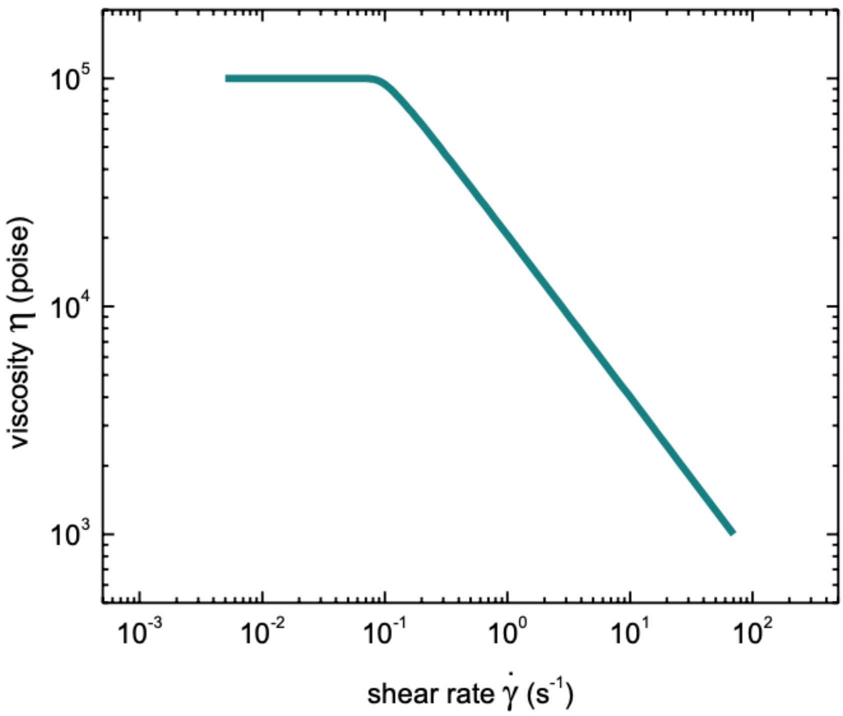
ChE 596 Polymer Rheology - Fall 2024 Homework 1

Due date: Tuesday, September 10th at 1:30 pm EDT

Show all your work for full credit.

Problem 1

As a rheologist for ACME chemical, you have been assigned the task of developing the processing conditions for a polymer for developing plastic parts. A plot of the viscosity versus shear rate for the polymer is below. Due to instrument limitations, you can only extrude with a shear rate of either 7 or 25 sec^{-1} and you can cool it to a solid in either 5 sec or 15 sec. Which shear rate should you extrude the polymer, and which cooling would you choose? Why?



Problem 2

The components of a quantity F can be written as $F_j = a_i b_j c_i d_j$. What is F_3?

Problem 3

Evaluate the following using index notation:

A) $\underline{v}\ \underline{w} \bullet \underline{z}$

B) $\underline{\nabla}\ \underline{v} : \underline{v}\ \underline{v}$

Problem 4

Your colleague has been able to calculate the stress tensor (below) for a specific shear flow.

$$\underline{\underline{T}} = \begin{bmatrix} 7 & 5 & 0 \\ 5 & 7 & 0 \\ 0 & 0 & 7 \end{bmatrix}$$

What can you say about the fluid?